

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTEPC402 - DATABASE MANAGEMENT SYSTEM

Time : 3-Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The ability to change the conceptual schema without affecting the external schemas or application program is known _____.a.Program Data Independence b. Physical Data Independencec.Logical Data Independence d. Data Abstraction
Which level of abstraction describe What data are actually stored?a.Physical b.Logical c.External d.View
- 2 The logical design, and the snapshot of the data at a given instant in time is known as?a.Instance & Relation b. Relation & Schema c. Domain & Schema d.Schema & Instance
- 3 Either all of its operations are executed or none is called _____.a.Consistency b.Isolation c.Atomicity d.Durability
- 4 If one attribute is determinant of second, which in turn is determinant of third, then the relation cannot be:a.Well-structured b. 1NF c. 2NF d. 3NF
- 5 After groups have been established, SQL applies predicates in the _____ clause, allowing aggregate functions to be used.a.Where b. Having c. Group by d. Limit
- 6 If ____, then $A \rightarrow B$ has trivial functional dependency.a.B is a subset of A b.A is a subset of B c.A is a subset of A d.B is a superset of A
- 7 In order to build a link between two tables, which key is used?a.Primary b.Foreign c.Composite d.Unique
- 8 What is TRUE about SAVEPOINT?a.Following the completion of a transaction, it must be executed to save all the operations performed in the transaction.b.A transaction can be rolled back to its last saved state.c.A specific part of a transaction can be given a named.None of the above
- 9 Which command will remove the records from the table, but not affect the structure of the table?a.Remove b.Delete c.Drop d.Truncate
- 10 What is the outcome of the command `db.createCollection('myCollection')` in MongoDB?a.Creates a new documentb.Deletes a collectionc.Creates a new collectiond.Updates a collection

12. What is a serial schedule?
 a. A schedule where transactions are interleaved.
 b. A schedule where transactions are executed one after another.
 c. A schedule with no conflicts.
 d. A schedule that is not recoverable.

Q2. Solve the following.

- A) Explain 1-tire, 2-tire, and 3-tire database architecture in detail. 6
 B) Differentiate between Traditional file processing system and DBMS. 6

Q3. Solve the following.

- Consider the following database Emp (emp_no, emp_name, salary, department, city). Write SQL statements for following:
 1. change employee name from "Rahul" to "Ganesh"
 2. delete the record for the employee who are from "Delhi"
 3. Out of all employees from "production" Department, list out only two employee details. 6
 Explain the aggregate functions in SQL with their syntax and example. 6

Q4. Solve Any Two of the following.

- A) Explain 2NF, 3NF and BCNF with examples. Explain why BCNF is more strict than 3NF? 6
 B) Explain the different functional dependencies used in DBMS. 6
 C) Explain candidate key, Super key and Foreign key. 6

Q5. Solve Any Two of the following.

- What is Serializability in scheduling? Explain conflict Serializability in detail. 6
 Explain lock and timestamp based concurrency control protocol. 6
 Explain ACID property in detail. 6

Q6. Solve Any Two of the following.

- A) State and provide a detailed explanation of CAP theorem in distributed data systems. 6
 B) Explain different Types of NOSQL database. 6
 C) What are the essential CRUD operations in MongoDB? Illustrate each briefly. 6

*** End ***